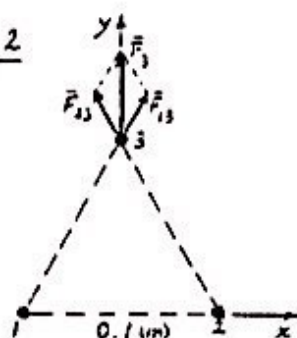


Chapter 3

Static Electric Fields

- P. 3-1 a) Max. voltage $V_{\max}$ will make $d_f = h/2$ at $z = w$.
- $$\frac{h}{2} = \frac{e}{2m} \left(\frac{V_{\max}}{h} \right) \left(\frac{w}{u_0} \right)^2 \rightarrow V_{\max} = \frac{m}{e} \left(\frac{u_0 h}{w} \right)^2.$$
- b) At the screen, $(d_0)_{\max} = D/2$. L must be $\leq L_{\max}$, where
- $$L_{\max} = \frac{1}{2} \left(w + \frac{m u_0^2 D h}{e w V_{\max}} \right).$$
- c) Double $V_{\max}$ by doubling u_0^2 , or doubling the anode accelerating voltage.

P. 3-2



$$\begin{aligned} \vec{F}_{13} &= \frac{(2 \times 10^{-6})^2}{4\pi\epsilon_0 (0.1)^2} (\vec{a}_x 0.5 + \vec{a}_y 0.866) \\ &= 3.6 (\vec{a}_x 0.5 + \vec{a}_y 0.866) \text{ (N)} \\ \vec{F}_{23} &= 3.6 (-\vec{a}_x 0.5 + \vec{a}_y 0.866) \text{ (N)} \\ \vec{F}_3 &= \vec{F}_{13} + \vec{F}_{23} = \vec{a}_y 0.624 \text{ (N)}. \end{aligned}$$

Similarly for $\vec{F}_1$ and $\vec{F}_2$. All are repulsive forces in the direction away from the center of the triangle.

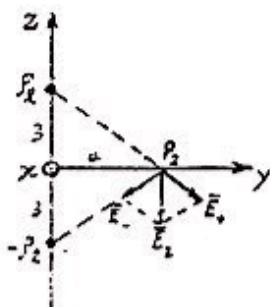
P. 3-3 $\vec{Q}_1 \cdot \vec{p} = -\vec{a}_y 3 + \vec{a}_z 4$, $\vec{Q}_2 \cdot \vec{p} = \vec{a}_y 4 - \vec{a}_z 3$.

At P: $\vec{E}_1 = \frac{Q_1}{4\pi\epsilon_0 (5)^3} (-\vec{a}_y 3 + \vec{a}_z 4)$.

$$\vec{E}_2 = \frac{Q_2}{4\pi\epsilon_0 (5)^3} (\vec{a}_y 4 - \vec{a}_z 3).$$

- a) No y-component: $-3Q_1 + 4Q_2 = 0 \rightarrow \frac{Q_1}{Q_2} = \frac{4}{3}$.
- b) No z-component: $4Q_1 - 3Q_2 = 0 \rightarrow \frac{Q_1}{Q_2} = \frac{3}{4}$.





b) At $P_2 (0, 4, 0)$ on the xy -plane (the ground):

Vector from P_1 to P_2 is $\vec{r}_+ = \vec{a}_y 4 - \vec{a}_z 4$

Vector from $-P_1$ to P_2 is $\vec{r}_- = \vec{a}_y 4 + \vec{a}_z 4$

$$\vec{E}_+ = \frac{\rho_l}{2\pi\epsilon_0} \frac{\vec{a}_y 4 - \vec{a}_z 4}{4^2 + 4^2}, \quad \vec{E}_- = \frac{-\rho_l}{2\pi\epsilon_0} \frac{\vec{a}_y 4 + \vec{a}_z 4}{4^2 + 4^2}$$

$$\vec{E}_2 = \vec{E}_+ + \vec{E}_- = \frac{\rho_l}{2\pi\epsilon_0} \left(\frac{-\vec{a}_z 8}{4^2 + 4^2} \right)$$

$$= 9 \times 10^5 (-\vec{a}_z 0.24) = -\vec{a}_z 2.16 \times 10^5 \text{ (V/m)}$$

$$\rho_{s2} = \epsilon_0 E_{2z} = \frac{\rho_l}{2\pi} (-0.24) = \frac{50 \times 10^{-6}}{2\pi} (-0.24) = -1.91 \times 10^{-6} \text{ (C/m}^2\text{)} \\ = -1.91 \text{ (}\mu\text{C/m}^2\text{)}$$

2.3-35 Given $D = 2 \text{ (cm)}$, $a = 0.3 \text{ (cm)}$.

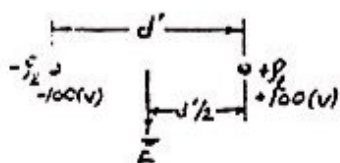
a) From Eq. (3-163),

$$d = \frac{1}{2} (D + \sqrt{D^2 - 4a^2}) = \frac{1}{2} [2 + \sqrt{2^2 - 4(0.3)^2}] = 1.954 \text{ (cm)}$$

$$d_i = D - d = 2 - 1.954 = 0.046 \text{ (cm)} = 0.46 \text{ (mm)}$$

$$b) \rho_l = \frac{2\pi\epsilon_0 V_1}{\ln(d/a)} = \frac{2\pi(8.85 \times 10^{-12}) \times 100}{\ln(1.954/0.3)} = 2.96 \times 10^{-9} \text{ (F/m)} \\ = 2.96 \text{ (nF/m)}$$

c) The equivalent line charges are separated by



$$d' = d - d_i = 1.954 - 0.046 \\ = 1.908 \text{ (cm)}$$

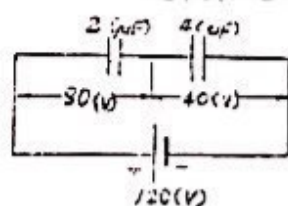
$$|\vec{E}| = \frac{\rho_l}{2\pi\epsilon_0 (d'/2)} \times 2 = 111.9 \text{ (V/m)}$$

— in a direction normal to the plane containing the wires.

P.3-24 Total capacitance across battery terminals,

$$C_T = \frac{4}{3} (\mu F).$$

Total stored electric energy $W_T = \frac{1}{2} C_T (120)^2 = 9.6 (\text{mJ}).$



$$W_e \text{ in } 2(\mu F) \text{ capacitor} = \frac{1}{2} (2 \times 10^{-6}) \times 80^2 = 6.4 (\text{mJ}).$$

$$W_e \text{ in } 4(\mu F) \text{ capacitor} = \frac{1}{2} (4 \times 10^{-6}) \times 40^2 = 0.8 (\text{mJ}).$$

$$W_e \text{ in } 3(\mu F) \text{ capacitor} = \frac{1}{2} (3 \times 10^{-6}) \times 40^2 = 2.4 (\text{mJ}).$$

P.3-25 $\vec{E} = \vec{a}_r 6r \sin \phi + \vec{a}_\phi 3r \cos \phi,$

$$d\vec{L} = \vec{a}_r dr + \vec{a}_\phi r d\phi + \vec{a}_z dz,$$

$$W_e = -q \int_P^B \vec{E} \cdot d\vec{L} = -(5 \times 10^{-10}) \left[6 \sin \phi \int_2^4 r dr + 3r^2 \int_{\pi/3}^{-\pi/2} \cos \phi d\phi \right].$$

a) First $r=2$, ϕ from $\pi/3$ to $-\pi/2$; then $\phi = -\pi/2$, r from 2 to 4:

$$W_e = -(5 \times 10^{-10}) \left[3(2)^2 \sin \phi \Big|_{\pi/3}^{-\pi/2} + 6 \sin \left(\frac{\pi}{2} \right) \frac{r^2}{2} \Big|_2^4 \right]$$

$$= -(5 \times 10^{-10}) [-18 - 36] = 27 \times 10^{-9} (\text{J}) = 27 (\text{nJ}).$$

b) First $\phi = \pi/3$, r from 2 to 4; then $r=4$, ϕ from $\pi/3$ to $-\pi/2$:

$$W_e = -(5 \times 10^{-10}) \left[6 \sin \left(\frac{\pi}{3} \right) \frac{r^2}{2} \Big|_2^4 + 3(4)^2 \sin \phi \Big|_{\pi/3}^{-\pi/2} \right]$$

$$= -(5 \times 10^{-10}) [18 - 72] = 27 \times 10^{-9} (\text{J}) = 27 (\text{nJ}).$$

— Same as W_e in part a). $\vec{\nabla} \times \vec{E} = 0 \rightarrow \vec{E}$ is conservative.

P.3-26 Assume the inner and outer radii to be a and $a+r$ respectively. Substituting Eq.(3-89) in Eq.(3-117) and using Eq.(3-115), we have

$$F_a = -\frac{\partial}{\partial r} \left(\frac{Q}{2} \cdot \frac{Q}{2\pi\epsilon L} \ln \frac{a+r}{a} \right)$$

$$= -\frac{Q^2}{4\pi\epsilon L(a+r)} = -\frac{Q^2}{4\pi\epsilon Lb}, \text{ in the direction of decreasing } r \text{ (attraction).}$$



Search



Download now



P.3-27 Switch open: Charges on the plates are Constant.

$$Q = CV_0, \quad W_e = \frac{Q^2}{2C}.$$

$$C = \frac{\epsilon_0 w}{d} [\epsilon x + \epsilon_0 (L-x)].$$

$$\begin{aligned} \bar{F}_e &= -\bar{\nabla} W_e = -\bar{a}_x \frac{Q^2}{2} \frac{\partial}{\partial x} \left(\frac{1}{C} \right) \\ &= \bar{a}_x \frac{Q^2 d}{2w} \frac{\epsilon - \epsilon_0}{[\epsilon x + \epsilon_0 (L-x)]^2} = \bar{a}_x \frac{V_0^2 w}{2d} (\epsilon - \epsilon_0). \end{aligned}$$

P.3-28 Use subscripts d and a to denote dielectric and air regions respectively. $\bar{\nabla}^2 V = 0$ in both regions.

$$V_d = c_1 y + c_2, \quad \bar{E}_d = -\bar{a}_y c_1, \quad \bar{D}_d = -\bar{a}_y \epsilon_0 \epsilon_r c_1.$$

$$V_a = c_3 y + c_4, \quad \bar{E}_a = -\bar{a}_y c_3, \quad \bar{D}_a = -\bar{a}_y \epsilon_0 c_3.$$

$$\text{B.C.: At } y=0, V_d = 0; \quad \text{at } y=d, V_a = V_0;$$

$$\text{at } y=0.8d: V_d = V_a, \quad \bar{D}_d = \bar{D}_a.$$

$$\text{Solving: } c_1 = \frac{V_0}{(0.8 + 0.2\epsilon_r)d}, \quad c_2 = 0, \quad c_3 = \frac{\epsilon_r V_0}{(0.8 + 0.2\epsilon_r)d}, \quad c_4 = \frac{(1 - \epsilon_r)V_0}{1 + 0.25\epsilon_r}.$$

$$a) V_d = \frac{5yV_0}{(4 + \epsilon_r)d}, \quad \bar{E}_d = -\bar{a}_y \frac{5V_0}{(4 + \epsilon_r)d}.$$

$$b) V_a = \frac{5\epsilon_r y - 4(\epsilon_r - 1)d}{(4 + \epsilon_r)d} V_0, \quad \bar{E}_a = -\bar{a}_y \frac{5\epsilon_r V_0}{(4 + \epsilon_r)d}.$$

$$c) (\rho_s)_{y=d} = -(D_a)_{y=d} = \frac{5\epsilon_r \epsilon_r V_0}{(4 + \epsilon_r)d}.$$

$$(\rho_s)_{y=0} = (D_d)_{y=0} = -\frac{5\epsilon_r \epsilon_r V_0}{(4 + \epsilon_r)d}.$$

P.3-29 Poisson's eq. $\bar{\nabla}^2 V = -\frac{A}{\epsilon r} \rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = -\frac{A}{\epsilon r}.$

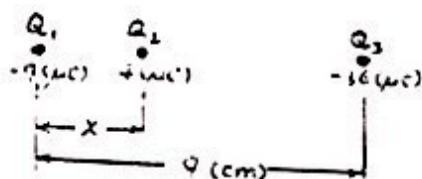
$$\text{Solution: } V = -\frac{A}{\epsilon} r + c_1 \ln r + c_2.$$

$$\text{B.C.: } \begin{cases} \text{At } r=a, V_0 = -\frac{A}{\epsilon} a + c_1 \ln a + c_2. \\ \text{At } r=b, 0 = -\frac{A}{\epsilon} b + c_1 \ln b + c_2. \end{cases}$$

$$c_1 = \frac{\frac{A}{\epsilon}(b-a) - V_0}{\ln(b/a)},$$

$$c_2 = \frac{V_0 \ln b + \frac{A}{\epsilon}(a \ln b - b \ln a)}{\ln(b/a)}.$$

P. 3-4



For zero force on Q_1 :

$$\frac{Q_1 Q_2}{4\pi\epsilon_0 x^2} + \frac{Q_1 Q_3}{4\pi\epsilon_0 9^2} = 0.$$

$$x = 9\sqrt{\frac{Q_2}{Q_3}} = 9\sqrt{\frac{4}{36}} = 3 \text{ (cm)}.$$

With $x = 3 \text{ (cm)}$, it can be proved that the net forces on Q_2 and Q_3 are also zero.

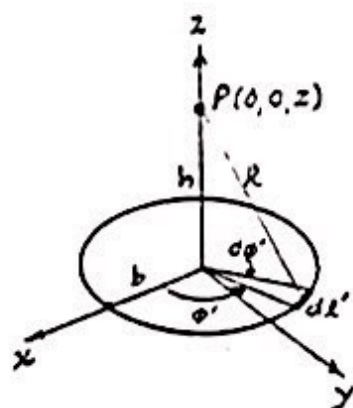
P. 3-5 From Eq. (3-42a), $\rho_s = \frac{\text{Total charge}}{\text{Disc area}} = \frac{Q}{\pi b^2}$.

$$\begin{aligned} \bar{E} &= \bar{a}_x \frac{\rho_s}{2\epsilon_0} \left[1 - \left(1 + \frac{b^2}{z^2} \right)^{-1/2} \right] \\ &= \bar{a}_x \frac{\rho_s}{2\epsilon_0} \left[1 - \left(1 - \frac{b^2}{2z^2} + \frac{3}{8} \frac{b^4}{z^4} - \dots \right) \right] \\ &= \bar{a}_x \frac{\rho_s}{2\pi\epsilon_0 b^2} \left[\frac{1}{2} \left(\frac{b}{z} \right)^2 - \frac{3}{8} \left(\frac{b}{z} \right)^4 + \dots \right] \\ &= \bar{a}_x \left[\frac{Q}{4\pi\epsilon_0 z^2} \left(1 - \frac{3}{4} \frac{b^2}{z^2} + \dots \right) \right], \end{aligned}$$

where the first term is the point-charge term and the rest represent the error. Considering only the first error term:

$$\frac{3}{4} \cdot \frac{b^2}{z^2} \leq 0.01 \rightarrow z \geq \sqrt{75} b, \text{ or } 8.66 b.$$

P. 3-6 At an arbitrary $P(0,0,z)$ on the axis:



$$dV_p = \frac{\rho_s b d\phi'}{4\pi\epsilon_0 (z^2 + b^2)^{1/2}}.$$

$$V_p = \frac{\rho_s b}{4\pi\epsilon_0 (z^2 + b^2)^{1/2}} \int_0^{2\pi} d\phi' = \frac{\rho_s b}{2\epsilon_0 (z^2 + b^2)^{1/2}}.$$

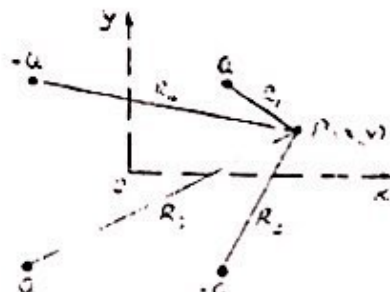
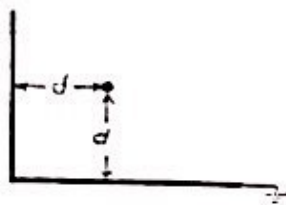
$$\bar{E}_p = -\bar{\nabla} V_p = -\bar{a}_z \frac{dV_p}{dz} = \bar{a}_z \frac{\rho_s b}{2\epsilon_0 (z^2 + b^2)^{3/2}}.$$

a) At point $(0,0,h)$, $\bar{E} = \bar{a}_z \frac{\rho_s b}{2\epsilon_0 (h^2 + b^2)^{3/2}}.$

b) To find the location of max. $|\bar{E}_p|$, set

$$\frac{\partial}{\partial z} |\bar{E}_p| = 0 \rightarrow z = \frac{b}{\sqrt{2}}. \quad \text{Max } |\bar{E}_p| = \frac{\rho_s}{2.67\epsilon_0 b}.$$

Similar situation when P is below the loop.



$$\begin{aligned}\vec{E}_P &= -\vec{\nabla} V_P = -\vec{a}_x \frac{\partial V_P}{\partial x} - \vec{a}_y \frac{\partial V_P}{\partial y} \\ &= \vec{a}_x \frac{Q}{4\pi\epsilon} \left[-\frac{x-d}{R_1^3} + \frac{x-d}{R_2^3} - \frac{x+d}{R_3^3} + \frac{x+d}{R_4^3} \right] \\ &\quad + \vec{a}_y \frac{Q}{4\pi\epsilon} \left[-\frac{y-d}{R_1^3} + \frac{y+d}{R_2^3} - \frac{y-d}{R_3^3} + \frac{y+d}{R_4^3} \right].\end{aligned}$$

E_P will have a z-component if the point P does not lie in the xy-plane.

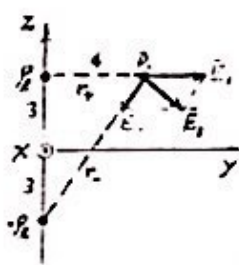
- b) On the conducting half-planes, $\rho_s = D_n = \epsilon E_n$.
Along the x-axis, $y=0$: $R_1 = [(x-d)^2 + d^2]^{1/2} = R_2$,
and $R_3 = [(x+d)^2 + d^2]^{1/2} = R_4$.

$$E_x = 0, \quad E_y = \frac{Q}{2\pi\epsilon} \left[\frac{d}{R_1^3} - \frac{d}{R_3^3} \right].$$

$$\begin{aligned}\therefore \rho_s(y=0) &= \frac{Qd}{2\pi} \left\{ \frac{1}{[(x-d)^2 + d^2]^{3/2}} - \frac{1}{[(x+d)^2 + d^2]^{3/2}} \right\} \\ &= \begin{cases} 0, & \text{at } x=0. \\ \text{max.}, & \text{at } x=d. \end{cases}\end{aligned}$$

Similarly for $\rho_s(x=0)$ on the vertical conducting plane by changing x to y .

P.3-34



Assume ρ_L (50 nC/m) to be at $y=0$ and $z=3$ (m)

a) Vector from P_2 to $P_1(0, 4, 3)$: $\vec{r}_1 = \vec{a}_y 4$

Vector from $-P_2$ to P_1 : $\vec{r}_2 = \vec{a}_y 4 + \vec{a}_z 6$

$$\vec{E}_+ = \frac{\rho_L}{2\pi\epsilon_0} \frac{\vec{r}_1}{r_1^3} = \frac{\rho_L}{2\pi\epsilon_0} \frac{\vec{a}_y}{4} = 9 \times 10^5 (\vec{a}_y 0.25)$$

$$\vec{E}_- = \frac{-\rho_L}{2\pi\epsilon_0} \frac{\vec{r}_2}{r_2^3} = -9 \times 10^5 (\vec{a}_y 0.077 + \vec{a}_z 0.115)$$

$$\vec{E}_1 = \vec{E}_+ + \vec{E}_- = 9 \times 10^5 (\vec{a}_y 0.173 - \vec{a}_z 0.115) \text{ (V/m) at } P_1$$

P.3-30 $\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = 0.$

Solution: $V = c_1 \ln r + c_2.$

Boundary conditions:

At $r = a$, $V = V_0 = c_1 \ln a + c_2.$

At $r = b$, $V = 0 = c_1 \ln b + c_2.$

$c_1 = -\frac{V_0}{\ln(b/a)}, \quad c_2 = \frac{V_0 \ln b}{\ln(b/a)}.$

$V = V_0 \frac{\ln(b/r)}{\ln(b/a)}, \quad \vec{E} = -\nabla V = -\vec{a}_r \frac{V_0}{r \ln(b/a)}.$

Surface densities: At $r = a$, $\rho_{sa} = \epsilon_0 E_r = \frac{\epsilon_0 V_0}{a \ln(b/a)}.$

At $r = b$, $\rho_{sb} = -\epsilon_0 E_r = -\frac{\epsilon_0 V_0}{b \ln(b/a)}.$

Capacitance per unit length $C' = \frac{Q}{V_{ab}} = \frac{2\pi a \rho_{sa}}{V_0} = \frac{2\pi \epsilon_0}{\ln(b/a)} \quad (\text{C/m}).$

P.3-31 V and $\vec{E}$ depend only on θ . $\rightarrow$ Eq (3-129): $\frac{d}{d\theta} \left(\sin \theta \frac{dV}{d\theta} \right) = 0.$

a) Solution: $\frac{dV}{d\theta} = \frac{C_1}{\sin \theta} \rightarrow V(\theta) = C_1 \ln \left(\tan \frac{\theta}{2} \right) + C_2.$

B.C. ① $V(\alpha) = V_0 = C_1 \ln \left(\tan \frac{\alpha}{2} \right) + C_2.$

② $V(\frac{\pi}{2}) = 0 = C_1 \ln \left(\tan \frac{\pi}{4} \right) + C_2 \rightarrow C_2 = 0.$

$C_1 = \frac{V_0}{\ln \left[\tan(\alpha/2) \right]} \rightarrow V(\theta) = \frac{V_0 \ln \left[\tan(\theta/2) \right]}{\ln \left[\tan(\alpha/2) \right]}.$

b) $\vec{E} = -\vec{a}_\theta \frac{dV}{R d\theta} = -\vec{a}_\theta \frac{V_0}{R \ln \left[\tan(\alpha/2) \right] \sin \theta}.$

c) On the cone: $\theta = \alpha$, $\rho_s = \epsilon_0 E(\alpha) = -\frac{\epsilon_0 V_0}{R \ln \left[\tan(\alpha/2) \right] \sin \alpha}.$

On the grounded plane: $\theta = \pi/2$, $\rho_s = -\epsilon_0 E(\frac{\pi}{2}) = \frac{\epsilon_0 V_0}{R \ln \left[\tan(\alpha/2) \right]}.$

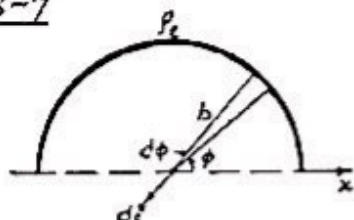
P.3-32 Consider the conditions in the xy -plane ($z=0$).

a) $V_0 = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3} - \frac{1}{R_4} \right)$, where

$R_1 = [(x-d)^2 + (y-d)^2]^{1/2}, \quad R_2 = [(x-d)^2 + (y+d)^2]^{1/2},$

$R_3 = [(x+d)^2 + (y-d)^2]^{1/2}, \quad R_4 = [(x+d)^2 + (y+d)^2]^{1/2}.$

P.3-7



$$dE_y = -\frac{\rho_e(b d\phi)}{4\pi\epsilon_0 b^2} \sin\phi,$$

$$\bar{E} = \bar{a}_y E_y = -\bar{a}_y \frac{\rho_e}{4\pi\epsilon_0 b} \int_0^\pi \sin\phi d\phi$$

$$= -\bar{a}_y \frac{\rho_e}{2\pi\epsilon_0 b}.$$

P.3-8 Spherical symmetry: $\bar{E} = \bar{a}_r E_r$. Apply Gauss's law.

$$1) 0 \leq R \leq b. \quad 4\pi R^2 E_{R1} = \frac{\rho_e}{\epsilon_0} \int_0^R \left(1 - \frac{R^2}{b^2}\right) 4\pi R^2 dR = \frac{4\pi\rho_e}{\epsilon_0} \left(\frac{R^3}{3} - \frac{R^5}{5b^2}\right),$$

$$E_{R1} = \frac{\rho_e}{\epsilon_0} R \left(\frac{1}{3} - \frac{R^3}{5b^2}\right).$$

$$2) b \geq R < R_i. \quad 4\pi R^2 E_{R2} = \frac{\rho_e}{\epsilon_0} \int_0^b \left(1 - \frac{R^2}{b^2}\right) 4\pi R^2 dR = \frac{8\pi\rho_e}{15\epsilon_0} b^3,$$

$$E_{R2} = \frac{2\rho_e b^3}{15\epsilon_0 R^2}.$$

$$3) R_i < R < R_o. \quad E_{R3} = 0.$$

$$4) R > R_o. \quad E_{R4} = \frac{2\rho_e b^3}{15\epsilon_0 R^2}.$$

P.3-9 Cylindrical symmetry: $\bar{E} = \bar{a}_r E_r$. Apply Gauss's law.

$$a) E_r = 0, \text{ for } r < a.$$

$$E_r = \frac{a\rho_{sa}}{\epsilon_0 r}, \text{ for } a < r < b.$$

$$E_r = \frac{a\rho_{sa} + b\rho_{sb}}{\epsilon_0 r}, \text{ for } r > b.$$

$$b) \frac{b}{a} = -\frac{\rho_{sa}}{\rho_{sb}}.$$

$$P.3-10 \quad W_e = -q \int \bar{E} \cdot d\bar{l} = -q \int (y dx + x dy).$$

$$a) \text{ Along the parabola } y = 2x^2, \quad dy = 4x dx.$$

$$W_e = -(5 \times 10^{-6}) \int_1^{-2} (2x^2 + 4x^2) dx = 9 \times 10^{-5} (J) = 90 (\mu J).$$

$$b) \text{ Along the straight line } \frac{y-2}{x-1} = \frac{8-2}{-2-1} = -2, \quad y = -2x + 4, \quad dy = -2dx.$$

$$W_e = -(5 \times 10^{-6}) \int_1^{-2} [(-2x + 4) dx - 2x dx] = 90 (\mu J).$$

P. 3-17 Use subscript a for air, p for plexiglass, and b for breakdown.

$$a) V_b = E_{ba} d_a = (3 \times 10^6) \times (50 \times 10^{-3}) = 150 \times 10^3 (V) = 150 (kV).$$

$$b) V_b = E_{bp} d_p = 20 \times 50 = 1,000 (kV).$$

$$c) V_b = E_a d_a + E_p d_p = E_a (50 - d_p) + E_p d_p$$

$$\text{Now } D_a = D_p \rightarrow \epsilon_0 E_a = \epsilon_0 \epsilon_{rp} E_p \rightarrow E_a = \epsilon_{rp} E_p > E_p.$$

$$E_{ba} < E_{bp} \rightarrow \text{Breakdown occurs in air region first.}$$

$$\therefore V_b = E_{ba} (50 - 10) + \frac{E_{ba}}{3} \times 10 = 3(40 - \frac{1}{3} \times 10) = 130 (kV).$$

P. 3-18 At the $z=0$ plane: $\vec{E}_1 = \vec{a}_x 2y - \vec{a}_y 3x + \vec{a}_z 5$.

$$\vec{E}_1(z=0) = \vec{E}_2(z=0) = \vec{a}_x 2y - \vec{a}_y 3x.$$

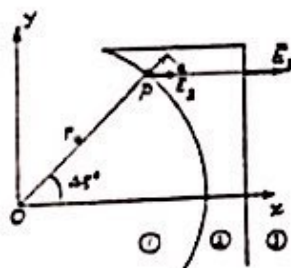
$$\vec{D}_{1n}(z=0) = \vec{D}_{2n}(z=0) \rightarrow 2\vec{E}_{1n}(z=0) = 3\vec{E}_{2n}(z=0),$$

$$\rightarrow E_{2n}(z=0) = \frac{2}{3}(\vec{a}_z 5) = \vec{a}_z \frac{10}{3}.$$

$$\therefore \vec{E}_2(z=0) = \vec{a}_x 2y - \vec{a}_y 3x + \vec{a}_z \frac{10}{3}.$$

$$\vec{D}_2(z=0) = (\vec{a}_x 2y - \vec{a}_y 3x + \vec{a}_z \frac{10}{3}) 3\epsilon_0$$

P. 3-19



$$\text{Assume } \vec{E}_3 = \vec{a}_r E_{3r} + \vec{a}_\phi E_{3\phi}.$$

$$\text{Boundary condition: } \vec{a}_n \cdot \vec{E}_1 = \vec{a}_n \cdot \vec{E}_2.$$

$$\rightarrow E_{2\phi} = -3.$$

$$\text{For } \vec{E}_3, \text{ and hence } \vec{E}_2, \text{ to be parallel to the } x\text{-axis, } E_{2\phi} = -E_{2r}.$$

$$\rightarrow E_{2r} = 3.$$

$$\text{Boundary condition: } \vec{a}_n \cdot \vec{D}_1 = \vec{a}_n \cdot \vec{D}_2.$$

$$\rightarrow \epsilon_1 E_{1r} = \epsilon_2 E_{2r} \rightarrow \epsilon_0 5 = \epsilon_0 \epsilon_{r2} 3.$$

$$\rightarrow \epsilon_{r2} = \frac{5}{3} = 1.667.$$

P.3-20 $\epsilon = \frac{\epsilon_2 - \epsilon_1}{d} y + \epsilon_1$.

Assume Q on plate at $y=d$. $\vec{E} = -\vec{a}_y \frac{\rho_s}{\epsilon} = \frac{Q}{s(\frac{\epsilon_2 - \epsilon_1}{d} y + \epsilon_1)}$.

$$V = -\int_{y=0}^{y=d} \vec{E} \cdot d\vec{l} = \frac{Qd \ln(\epsilon_2/\epsilon_1)}{s(\epsilon_2 - \epsilon_1)}.$$

$$C = \frac{Q}{V} = \frac{s(\epsilon_2 - \epsilon_1)}{d \ln(\epsilon_2/\epsilon_1)}.$$

P.3-21 Let ρ_ℓ be the lineal charge density on the inner conductor. $\vec{E} = \vec{a}_r \frac{\rho_\ell}{2\pi\epsilon r}$.

$$V_0 = -\int_b^a \vec{E} \cdot d\vec{r} = \frac{\rho_\ell}{2\pi\epsilon} \ln\left(\frac{b}{a}\right) \rightarrow \rho_\ell = \frac{2\pi\epsilon V_0}{\ln(b/a)}.$$

a) $\vec{E}(a) = \vec{a}_r \frac{V_0}{a \ln(b/a)}$.

b) For a fixed b , the function to be minimized is: $(x = b/a)$.

$$f(x) = \frac{V_0 x}{b \ln x}. \text{ Setting } \frac{df(x)}{dx} = 0 \text{ yields } \ln x = 1,$$

$$\text{or } x = \frac{b}{a} = e = 2.718.$$

c) $\min. E(a) = e V_0 / b$.

d) $C' = \frac{\rho_\ell}{V_0} = \frac{2\pi\epsilon}{\ln(b/a)} = 2\pi\epsilon \text{ (F/m)}$

P.3-22 $\vec{D} = \vec{a}_r \frac{\rho_\ell}{2\pi r}$. $\vec{E}_1 = \vec{a}_r \frac{\rho_\ell}{2\pi\epsilon_0\epsilon_{r1}r}$, $r_i < r < b$;

$$\vec{E}_2 = \vec{a}_r \frac{\rho_\ell}{2\pi\epsilon_0\epsilon_{r2}r}, \quad b < r < r_o$$

$$V = -\int_{r_o}^{r_i} \vec{E} \cdot d\vec{r} = \frac{\rho_\ell}{2\pi\epsilon_0} \left[\frac{1}{\epsilon_{r1}} \ln\left(\frac{b}{r_i}\right) + \frac{1}{\epsilon_{r2}} \ln\left(\frac{r_o}{b}\right) \right],$$

$$C' = \frac{\rho_\ell}{V} = \frac{2\pi\epsilon_0}{\frac{1}{\epsilon_{r1}} \ln\left(\frac{b}{r_i}\right) + \frac{1}{\epsilon_{r2}} \ln\left(\frac{r_o}{b}\right)} \text{ (F/m)}.$$

P.3-23 Assume charges $+Q$ and $-Q$ on the inner and outer conductors, respectively. $\vec{E} = \vec{a}_R E_R = \vec{a}_R \frac{Q}{4\pi\epsilon_0 R^2}$.

$$V = -\int_{R_o}^{R_i} \vec{E} \cdot \vec{a}_R dR = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_i} - \frac{1}{R_o} \right).$$

$$C = \frac{Q}{V} = \frac{4\pi\epsilon}{(1/R_i - 1/R_o)}.$$

P.3-15 $\vec{P} = P_0 (\vec{a}_x 3x + \vec{a}_y 4y)$.

a) $\rho_{pv} = -\vec{\nabla} \cdot \vec{P} = -7P_0$.

Total volume charge $Q_v = -7P_0 \pi (r_o^2 - r_i^2)$ per unit length.

Outer surface: $r = r_o$, $\vec{a}_n = \vec{a}_r$. $\rho_{ps} = \vec{P} \cdot \vec{a}_r = P_0 (\vec{a}_x 3r_o \cos \phi + \vec{a}_y 4r_o \sin \phi) \cdot \vec{a}_r$
 $= P_0 r_o (3 \cos^2 \phi + 4 \sin^2 \phi)$
 $= P_0 r_o (3 + \sin^2 \phi)$.

Inner surface: $r = r_i$, $\vec{a}_n = -\vec{a}_r$. $\rho_{ps} = -P_0 r_i (3 + \sin^2 \phi)$.

b) Total $Q_{so} = \int_0^{2\pi} \rho_{ps} r_o d\phi = P_0 r_o^2 \int_0^{2\pi} (3 + \sin^2 \phi) d\phi = 7\pi P_0 r_o^2$, per unit length.

Total $Q_{si} = -7\pi P_0 r_i^2$, per unit length.

Total bound charge: $Q_v + Q_{so} + Q_{si} = 0$.

P.3-16 Spherical symmetry: Apply Gauss's law. $\vec{E} = \vec{a}_R E_R$, $\vec{D} = \vec{a}_R D_R$.

(1) $R > R_o$. $E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2}$, $V_1 = \frac{Q}{4\pi\epsilon_0 R}$.

$D_{R1} = \epsilon_0 E_{R1} = \frac{Q}{4\pi R^2}$, $P_{R1} = 0$.

(2) $R_i < R < R_o$. $E_{R2} = \frac{Q}{4\pi\epsilon_0 \epsilon_r R^2}$, $D_{R2} = \frac{Q}{4\pi R^2}$.

$P_{R2} = D_{R2} - \epsilon_0 E_{R2} = \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R^2}$.

$V_2 = -\int_{\infty}^{R_o} E_{R1} dR - \int_{R_o}^R E_{R2} dR = \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} - \frac{1}{\epsilon_r R} \right]$.

(3) $R < R_i$. $E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2}$, $D_{R3} = \frac{Q}{4\pi R^2}$, $P_{R3} = 0$.

$V_3 = V_2 \Big|_{R=R_i} - \int_{R_i}^R E_{R3} dR$
 $= \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} - \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_i} + \frac{1}{R} \right]$.

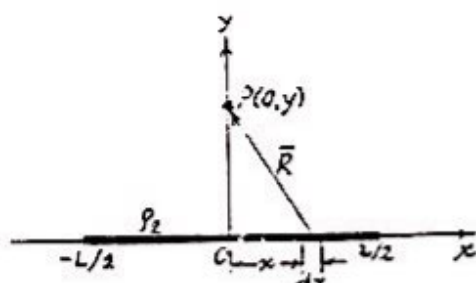
P.3-11 $\vec{E} = \bar{a}_x y - \bar{a}_y x$, $\vec{E} \cdot d\vec{l} = y dx - x dy$.

a) $W_e = -q \int_1^{-2} (2x^2 - 4x^3) dx = -30 (\mu J)$.

b) $W_e = -q \int_1^{-2} [(-2x+4)+2x] dx = -60 (\mu J)$.

The given $\vec{E}$ field is nonconservative.

P.3-12



a)
$$V = 2 \int_0^{L/2} \frac{\rho_0 dx}{4\pi\epsilon_0 R}$$

$$= \frac{\rho_0}{2\pi\epsilon_0} \int_0^{L/2} \frac{dx}{\sqrt{x^2 + y^2}}$$

$$= \frac{\rho_0}{2\pi\epsilon_0} \left\{ \ln \left[\sqrt{\left(\frac{L}{2}\right)^2 + y^2} + \frac{L}{2} \right] - \ln y \right\}.$$

b) From Coulomb's law:

$$\vec{E} = \bar{a}_y E_y = 2 \int_0^{L/2} \bar{a}_y \frac{\rho_0 y dx}{4\pi\epsilon_0 R^3} = \bar{a}_y \frac{\rho_0}{2\pi\epsilon_0 y} \left[\frac{L/2}{\sqrt{(L/2)^2 + y^2}} \right].$$

c) $\vec{E} = -\bar{\nabla} V$ gives the same answer as in b).

P.3-13 a) $\rho_{ps} = \bar{P} \cdot \bar{a}_n = P_0 \frac{L}{2}$ on all six faces of the cube.

$$\rho_{pv} = -\bar{\nabla} \cdot \bar{P} = -3P_0.$$

b) $Q_s = (6L^2)\rho_{ps} = 3P_0 L^3$, $Q_v = (L^3)\rho_{pv} = -3P_0 L^3$.

Total bound charge = $Q_s + Q_v = 0$.

P.3-14 $\bar{P} = \bar{a}_x P_0$.

a) $\rho_{ps} = \bar{P} \cdot \bar{a}_n = P_0 \sin\theta \cos\phi$.

$$\rho_{pv} = -\bar{\nabla} \cdot \bar{P} = 0.$$

b) $Q_s = \int_0^\pi \int_0^{2\pi} P_0 b^2 \sin^2\theta \cos\phi d\phi d\theta$
 $= 0.$